Bilateral Tail Dynamics and the Put–Call IV Ratio

How IV Regimes Govern Forward Return Distributions After Large Shocks

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Abstract

We study the role of the Put–Call implied volatility (IV) ratio not as a directional predictor, but as a state variable governing the geometry of conditional forward return distributions following large price shocks. Conditioning on shock direction (top or bottom 20% daily returns) and IV regime (puts expensive vs. calls expensive), we measure both lower and upper tails of cumulative forward returns over horizons of 1–20 trading days using Value at Risk, Expected Shortfall, and kurtosis.

Across assets (TSLA, MSFT, SPY, MSTR), we find that the IV ratio does not predict return direction. Instead, it selects whether forward distributions resolve through bilateral explosive tails, asymmetric recovery, or bounded near-Gaussian behavior. In convex single-names such as TSLA, put-dominated regimes following downside shocks produce extreme bilateral tail risk, with simultaneous crash and rally potential on the order of ±35–40% at 20-day horizons and pronounced kurtosis dynamics. In contrast, mega-cap equities and broad indices exhibit structural suppression of sustained bilateral explosiveness, even when short-horizon kurtosis spikes are extreme. Leveraged proxy exposure (MSTR) displays persistent bilateral explosiveness in all supported regimes, while certain IV regime–shock combinations are structurally absent, revealing constraints in option market pricing.

Kurtosis timing discriminates mechanisms: sustained elevation signals persistent fat-tailed risk, whereas sharp spikes followed by rapid normalization reflect transient dispersion and benign resolution. These findings demonstrate that measuring both tails is essential: one-sided risk metrics systematically mischaracterize uncertainty in regimes where bilateral tail risk dominates.

Overall, the Put–Call IV ratio functions as a selector of conditional return geometry, while asset structure determines whether bilateral convexity can manifest, remain bounded, or be structurally suppressed.

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1. Introduction

This study asks: Does the Put–Call IV ratio predict direction, or does it govern **the shape of forward return distributions** after large shocks?

By measuring **both tails** (5th and 95th percentiles) of forward returns conditional on shock direction and IV regime, we document that:

1. The IV ratio **does not predict sign**
2. It selects which scenarios exhibit **bilateral explosive tails** versus **asymmetric tail structures**
3. **Kurtosis reveals the mechanism:** high kurtosis signals fat bilateral tails
4. **Asset structure is destiny:** indices suppress bilateral explosiveness, single-names amplify it

Our most dramatic finding: **TSLA with puts expensive (IV>1) after a down shock exhibits -38% downside tail AND +36% upside tail simultaneously** (lag 20), with initial kurtosis of 8.5. This is massive bilateral uncertainty, not directional drift.

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2. Methodology

2.1 Four Scenarios

We condition on:

1. **Initial shock:** Large UP (top 20%) vs. Large DOWN (bottom 20%)
2. **IV ratio regime:** Puts expensive (IV > 1.05) vs. Calls expensive (IV < 0.95)

This yields four scenarios:

- **RevUp:** IV > 1 (puts expensive) + UP shock
- **ContDn:** IV > 1 (puts expensive) + DOWN shock
- **RevDn:** IV < 1 (calls expensive) + DOWN shock
- **ContUp:** IV < 1 (calls expensive) + UP shock

Labels "Rev" and "Cont" are code artifacts referring to IV regime, **not** to whether returns continue or reverse.

2.2 Bilateral Tail Statistics

For each scenario, we measure **BOTH tails** of the forward return distribution:

- **Lower tail (5%):** VaR_5 (5th percentile), ES_5 (expected shortfall in lower tail)
- **Upper tail (95%):** VaR_95 (95th percentile), ES_95 (expected shortfall in upper tail)
- **Kurtosis:** Fourth standardized moment measuring tail heaviness

All VaR and ES statistics are computed on cumulative forward log returns over horizon k, not on single-day returns; horizons therefore reflect distributional evolution rather than daily tail risk.

This bilateral approach is critical: measuring only one tail would miss that some scenarios exhibit **simultaneous explosive upside AND downside** (bilateral fat tails), while others show **asymmetric tail structures** (one fat tail, one thin tail).

2.3 Data

- **Assets:** TSLA, MSFT, SPY, MSTR
- **Sample period:** ~600 days lookback
- **IV data:** OVVO Labs 30-day maturity, 10 strikes around ATM
- **Forward horizons:** 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 trading days

2.4 Interpretation Framework

Bilateral explosive tails: ES_5 deeply negative AND ES_95 highly positive, with kurtosis > 5 - Indicates: Extreme uncertainty in both directions, fat-tailed distribution

Asymmetric tails: One tail explosive, other tail moderate - Indicates: Directional bias with one-sided tail risk

Thin bilateral tails: Both tails moderate, kurtosis < 3 - Indicates: Drift-dominated, near-Gaussian distribution

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3. Core Empirical Findings

3.1 TSLA — Bilateral Explosive Tails When Puts Are Expensive

TSLA exhibits stark regime-dependent differences in forward return geometry following large shocks. The Put–Call IV ratio does not predict the sign of future returns, but instead governs whether the forward

distribution exhibits bilateral explosive tails, asymmetric tail dominance, or thin-tailed drift behavior.

Table 1. TSLA ContDn (IV > 1, DOWN shock): BILATERAL EXPLOSIVENESS

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis	Interpretation
1	87	**-8.9%**	**+9.0%**	**8.55**	Extreme bilateral fat tails
5	86	**-17.1%**	**+18.8%**	3.90	Sustained bilateral tail risk
10	85	**-21.1%**	**+21.5%**	2.43	Kurtosis compresses, tails remain large
15	84	**-28.1%**	**+29.9%**	3.05	Bilateral expansion continues
20	83	**-38.4%**	**+36.2%**	4.06	**Massive bilateral tail risk**

Pattern. Following a large downside shock in a put-dominated IV regime, TSLA's forward return distribution exhibits extreme bilateral tail activation. At short horizons, kurtosis spikes to 8.55, indicating dominance by rare, extreme realizations in both directions. As the horizon extends, Expected Shortfall in both tails grows monotonically, reaching -38.4% on the downside and +36.2% on the upside by 20 trading days. Kurtosis compresses mid-horizon before re-elevating at longer horizons.

Interpretation. This regime represents **maximum uncertainty**, not directional bias. When TSLA falls and puts are expensive, the forward distribution becomes explosively two-sided: the asset may crash further or rally violently, with large tail mass in both directions simultaneously. This is neither mean reversion nor continuation, but genuine bilateral fat-tailed risk.

Economic meaning. Elevated put pricing in this regime does not signal downside dominance. Instead, it reflects fear of extreme outcomes in either direction. The IV ratio identifies a state of bilateral volatility explosion, not a bearish forecast.

Table 2. TSLA RevDn (IV < 1, DOWN shock): Asymmetric, Drift-Dominated with Upside Bias

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	19	-4.3%	**+3.8%**	1.62
10	19	-21.5%	**+12.5%**	2.43
20	19	**-16.0%**	**+22.3%**	2.21

Pattern. In the call-dominated IV regime following a downside shock, TSLA exhibits asymmetric tail behavior. Both tails are active, but the upper tail dominates economically. Kurtosis remains below Gaussian throughout, indicating thin-tailed dynamics.

Interpretation. When TSLA falls and calls are expensive, downside risk persists, but upside potential is larger. The absence of elevated kurtosis indicates that tail outcomes are generated through drift accumulation rather than convex amplification. This regime is fundamentally different from the bilateral explosiveness observed under put-dominated conditions.

Table 3. TSLA RevUp (IV > 1, UP shock): Thin-Tailed Downside Unwind (Rare)

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	5	-0.4%	+5.9%	2.04
20	5	**-23.9%**	+5.8%	0.80

Pattern. This scenario is extremely rare. The lower tail deepens substantially at longer horizons, while the upper tail remains modest. Kurtosis is persistently sub-Gaussian.

Interpretation. When TSLA rallies and puts become expensive, subsequent downside risk reflects an orderly unwind rather than crash dynamics. Losses accumulate through drift with thin tails, not through extreme events.

Table 4. TSLA ContUp (IV < 1, UP shock): Bilateral Expansion with Upside Bias

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	104	**-7.3%**	**+10.0%**	3.15
10	102	**-21.1%**	**+24.2%**	2.53
20	100	**-24.2%**	**+35.5%**	2.91

Pattern. Both tails widen steadily with horizon. Expected Shortfall reaches -24.2% on the downside and +35.5% on the upside by 20 trading days. Kurtosis remains near Gaussian throughout.

Interpretation. In this regime, TSLA exhibits bilateral risk with a clear upside bias. Tail expansion is drift-dominated rather than convex, consistent with momentum continuation accompanied by substantial two-way uncertainty.

TSLA Summary

Scenario	IV Regime	Shock	N	ES_5 (20d)	ES_95 (20d)	Kurtosis	Tail Structure
ContDn	IV > 1	DOWN	83	**-38.4%**	**+36.2%**	4.06–8.55	**Bilateral explosive**
RevDn	IV < 1	DOWN	19	-16.0%	+22.3%	1.6–2.4	Asymmetric, upside-biased
RevUp	IV > 1	UP	5	-23.9%	+5.8%	0.8–2.0	Thin-tailed unwind (rare)
ContUp	IV < 1	UP	100	-24.2%	**+35.5%**	2.5–3.2	Bilateral, drift-dominated

Core TSLA finding. The most dramatic regime is **ContDn**: when TSLA experiences a large downside shock and puts are expensive, the forward return distribution exhibits **simultaneous -38% downside risk and +36% upside potential**, with extreme short-horizon kurtosis. This is bilateral explosive uncertainty, not directional mean reversion or continuation.

3.2 SPY — Transient Tail Spikes and Structural Suppression of Bilateral Explosiveness

Unlike convex single-name equities, SPY exhibits strong structural suppression of sustained bilateral tail risk. Even when large shocks coincide with extreme IV regimes, tail behavior remains bounded, symmetric, and short-lived. The Put–Call IV ratio governs short-horizon dispersion but does not induce persistent convexity in index returns.

Table 5. SPY ContDn (IV > 1, DOWN shock): Benign Bilateral Tails

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	22	-1.0%	**+1.4%**	2.06
10	22	-4.1%	**+6.3%**	2.43
20	22	-11.4%	**+12.1%**	3.58

Pattern. Both tails remain modest across all horizons. By 20 trading days, downside Expected Shortfall reaches -11.4% while the upside tail reaches +12.1%. Kurtosis remains near Gaussian throughout, with no evidence of extreme tail thickening.

Interpretation. When SPY falls and puts are expensive, the forward return distribution remains balanced and well-behaved. This regime represents the most stable downside-shock scenario across all assets studied. Index diversification prevents bilateral tail explosion even when downside insurance is heavily demanded.

Table 6. SPY RevDn (IV < 1, DOWN shock): Extreme Initial Spike, Rapid Normalization

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	39	**-3.7%**	**+6.3%**	**14.22**
2	39	-4.3%	+4.8%	5.94
10	39	-6.3%	+8.5%	2.56
20	38	**-10.9%**	**+22.8%**	2.78

Pattern. This regime exhibits the single largest short-horizon kurtosis spike observed in the study, with kurtosis reaching 14.22 at lag 1. However, this spike collapses rapidly: by lag 10 kurtosis has normalized to near-Gaussian levels. At longer horizons, both tails are active but remain economically moderate.

Interpretation. When SPY falls and calls are expensive, the initial forward distribution reflects extreme one-day dispersion across index constituents, producing a sharp but transient fat-tail event. Rapid diversification-driven convergence follows. By 20 trading days, downside risk is limited (-10.9%) while the upside tail (+22.8%) reflects broad recovery rather than convex rebound. This is classic index behavior: initial chaos followed by rapid normalization.

Table 7. SPY RevUp (IV > 1, UP shock): Moderate Bilateral Expansion

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	35	-1.8%	+1.7%	4.24
20	33	-8.3%	+16.8%	2.48

Pattern. Both tails widen modestly with horizon. The upside tail exceeds the downside by lag 20, while kurtosis remains moderate.

Interpretation. When SPY rallies and puts are expensive, subsequent return distributions reflect limited two-way risk with mild upside bias. No convex tail amplification is observed.

Table 8. SPY ContUp (IV < 1, UP shock): Asymmetric but Bounded

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	29	-1.0%	**+1.6%**	3.78
20	29	-13.4%	**+10.6%**	3.38

Pattern. Both tails remain moderate. By lag 20, downside risk slightly exceeds upside potential. Kurtosis remains near Gaussian.

Interpretation. In this regime, SPY exhibits bounded bilateral risk with mild downside dominance. Distributional shape remains stable and thin-tailed, consistent with index-level diffusion rather than convex dynamics.

SPY Summary

Scenario	IV Regime	Shock	N	ES_5 (20d)	ES_95 (20d)	Kurtosis Pattern	Tail Structure
ContDn	IV > 1	DOWN	22	-11.4%	**+12.1%**	2.1–3.6 (stable)	Benign bilateral
RevDn	IV < 1	DOWN	38	-10.9%	**+22.8%**	**14.22 → 2.78**	Transient spike, rapid normalization
RevUp	IV > 1	UP	33	-8.3%	+16.8%	2.5–4.2	Moderate bilateral
ContUp	IV < 1	UP	29	-13.4%	+10.6%	3.4–3.8	Bounded, near-Gaussian

Core SPY finding. Even under extreme IV regimes and large shocks, SPY does not exhibit sustained bilateral explosive tails. The index structure suppresses convexity: tail spikes, when they occur, are short-lived and normalize quickly. Compared to TSLA's put-dominated downside regime (-38%/+36%, kurtosis > 8), SPY's worst-case scenarios remain economically and statistically bounded.

3.3 MSFT — Bounded Bilateral Tail Behavior in a Mega-Cap Equity

MSFT exhibits materially different tail dynamics from convex single-name equities such as TSLA. While short-horizon kurtosis spikes do occur under certain IV regimes, these effects are transient and resolve into bounded, near-Gaussian distributions at longer horizons. Across all scenarios, MSFT's forward return tails remain economically moderate, reflecting diversification, scale, and lower embedded leverage.

Table 9. MSFT ContDn (IV > 1, DOWN shock): Bounded Bilateral Tails

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	72	-8.7%	**+2.0%**	2.76
10	71	-7.6%	**+7.6%**	2.57
20	69	**-11.4%**	**+12.1%**	3.58

Pattern. Following a large downside shock in a put-dominated IV regime, MSFT's forward return distribution remains well balanced. By 20 trading days, downside and upside Expected Shortfall are nearly symmetric (-11.4% versus +12.1%). Kurtosis remains close to Gaussian throughout.

Interpretation. When MSFT falls and puts are expensive, bilateral tail risk is present but bounded. Losses and gains accumulate gradually through drift rather than convex amplification. The absence of elevated kurtosis indicates stable distributional structure rather than explosive tail behavior.

Table 10. MSFT RevDn (IV < 1, DOWN shock): Transient Spike, Bounded Resolution

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	36	**-5.0%**	**+10.8%**	**8.44**
2	36	-5.3%	+6.2%	3.53
10	35	-7.0%	+11.6%	4.76
20	35	**-10.4%**	**+22.8%**	2.62

Pattern. This regime exhibits a pronounced but short-lived kurtosis spike at the one-day horizon (K = 8.44), followed by rapid compression. By longer horizons, kurtosis normalizes toward Gaussian levels while tail magnitudes remain moderate. Upside tail risk increasingly dominates at longer horizons.

Interpretation. When MSFT falls and calls are expensive, the initial forward distribution reflects short-horizon dispersion across large-cap constituents rather than sustained convexity. While upside recovery potential becomes economically larger than downside risk by 20 days, the distribution remains thin-tailed. This behavior mirrors index-like dynamics more than single-name convexity.

Table 11. MSFT RevUp (IV > 1, UP shock): Moderate Bilateral Expansion

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	27	-1.9%	**+2.3%**	3.82
20	27	**-9.9%**	**+20.6%**	4.47

Pattern. Both tails widen with horizon, with the upside tail exceeding the downside by 20 days. Kurtosis rises modestly but does not indicate explosive tail behavior.

Interpretation. When MSFT rallies and puts are expensive, forward returns reflect moderate two-way risk with upside dominance. Tail expansion remains controlled, consistent with gradual rebalancing rather than convex amplification.

Table 12. MSFT ContUp (IV < 1, UP shock): Bounded Bilateral with Mild Downside Bias

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	78	-2.3%	**+1.8%**	3.82
20	73	**-13.4%**	**+10.6%**	3.33

Pattern. Both tails expand moderately with horizon. Downside risk slightly exceeds upside potential at longer horizons. Kurtosis remains near Gaussian.

Interpretation. In this regime, MSFT exhibits stable, bounded bilateral risk with mild downside dominance. Distributional shape remains diffusive and well behaved.

MSFT Summary

Scenario	IV Regime	Shock	N	ES_5 (20d)	ES_95 (20d)	Kurtosis Pattern	Tail Structure
ContDn	IV > 1	DOWN	69	-11.4%	**+12.1%**	2.6–3.6	Bounded bilateral
RevDn	IV < 1	DOWN	35	-10.4%	**+22.8%**	**8.44 → 2.62**	Transient spike, upside recovery
RevUp	IV > 1	UP	27	-9.9%	+20.6%	3.8–4.5	Moderate bilateral
ContUp	IV < 1	UP	73	-13.4%	+10.6%	2.5–3.4	Bounded bilateral

Core MSFT finding. MSFT exhibits bounded bilateral tail behavior across all regimes. Short-horizon kurtosis spikes can occur, particularly following downside shocks in call-dominated IV regimes, but these effects normalize rapidly. Terminal tail magnitudes remain far smaller than those observed in convex single-name equities, reflecting the stabilizing influence of scale, diversification, and lower embedded leverage.

3.4 MSTR — Structural Bilateral Explosiveness in Call-Dominated Regimes

MSTR differs fundamentally from all other assets studied. Under the IV-ratio construction used here, put-dominated regimes (IV > 1) do not occur in-sample. As a result, MSTR does not transition between IV regimes in the same way as TSLA, MSFT, or SPY. Instead, where observations exist, MSTR exhibits persistent and extreme bilateral tail risk driven by its embedded leveraged exposure to Bitcoin.

Table 11. MSTR RevDn (IV < 1, DOWN shock): BILATERAL EXPLOSIVE

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	120	**-12.6%**	**+15.5%**	4.05
10	118	**-27.6%**	**+41.4%**	3.42
20	116	**-32.2%**	**+70.2%**	4.04

Pattern. Following a large downside shock in a call-dominated IV regime, MSTR's forward return distribution exhibits extreme bilateral tail expansion. Downside Expected Shortfall deepens from -12.6% to -32.2% by 20 trading days, while the upper tail expands dramatically from +15.5% to +70.2%. Kurtosis remains persistently elevated.

Interpretation. When MSTR falls and calls are expensive, the forward distribution becomes explosively two-sided. Large additional drawdowns coexist with extraordinary rebound potential. This is not mean reversion or drift-driven recovery; it reflects structural convexity embedded in the asset's payoff profile. Both tails amplify simultaneously, producing sustained bilateral fat-tailed risk.

Table 12. MSTR ContUp (IV < 1, UP shock): BILATERAL EXPLOSIVE

Lag	N	ES_5 (Lower)	ES_95 (Upper)	Kurtosis
1	120	**-14.9%**	**+13.6%**	4.46
10	118	**-27.5%**	**+59.0%**	4.02
20	116	**-34.8%**	**+82.5%**	3.26

Pattern. Even following large upside shocks, MSTR exhibits extreme bilateral tail risk. By 20 trading days, downside Expected Shortfall reaches -34.8% while the upside tail expands to +82.5%. Kurtosis remains elevated throughout.

Interpretation. When MSTR rallies and calls remain expensive, the forward distribution continues to display massive two-sided risk. The coexistence of severe crash risk and extraordinary upside potential reflects leveraged exposure to the underlying crypto asset. Bilateral explosiveness persists regardless of initial shock direction.

Table 13. MSTR RevUp and ContDn: STRUCTURALLY UNSUPPORTED REGIMES

- **RevUp (IV > 1, UP shock):** N = 0 across all horizons
- **ContDn (IV > 1, DOWN shock):** N = 0 across all horizons

Interpretation. Under the IV-ratio thresholds used in this study, MSTR does not enter put-dominated regimes. This absence does not indicate a lack of downside risk. Instead, it reflects the option market's structural pricing of MSTR as a call-like, crypto-levered instrument. Downside risk is embedded directly in the underlying exposure rather than being expressed through relatively expensive puts. The nonexistence of IV > 1 regimes is itself a defining feature of MSTR's return geometry.

MSTR Summary

Scenario	IV Regime	Shock	N	ES_5 (20d)	ES_95 (20d)	Kurtosis	Tail Structure
RevUp	IV > 1	UP	0	—	—	—	Structurally absent
ContDn	IV > 1	DOWN	0	—	—	—	Structurally absent
RevDn	IV < 1	DOWN	116	**-32.2%**	**+70.2%**	3.4–4.1	Bilateral explosive
ContUp	IV < 1	UP	116	**-34.8%**	**+82.5%**	3.3–4.5	Bilateral explosive

Core MSTR finding. Where observations exist, MSTR exhibits extreme bilateral explosive tail behavior independent of shock direction. Unlike other assets, MSTR does not transition between IV regimes; instead, it remains locked in a call-dominated state with persistent two-sided convexity. The leveraged Bitcoin exposure

creates simultaneous crash risk and extraordinary upside potential, producing the most extreme bilateral tail geometry observed in the study.

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4. Cross-Asset Synthesis

This section synthesizes the lag-20 bilateral tail results across assets to highlight how IV regimes and asset structure jointly govern forward return geometry. All scenario labels follow the global definitions: ContDn denotes IV > 1 with a DOWN shock; RevDn denotes IV < 1 with a DOWN shock; ContUp denotes IV < 1 with an UP shock. No directional interpretation is implied by the labels.

Table 14. Bilateral Tail Comparison: Lag 20 Results

Asset	Scenario	N	ES_5	ES_95	Kurtosis	Tail Character
TSLA	**ContDn (IV > 1, DOWN)**	83	**-38.4%**	**+36.2%**	4.06	**Bilateral explosive**
TSLA	**RevDn (IV < 1, DOWN)**	19	-16.0%	+22.3%	2.21	Asymmetric, upside-biased
TSLA	**ContUp (IV < 1, UP)**	100	-24.2%	**+35.5%**	2.91	Bilateral, upside-biased
SPY	**ContDn (IV > 1, DOWN)**	22	-11.4%	+12.1%	3.58	Benign bilateral
SPY	**RevDn (IV < 1, DOWN)**	38	-10.9%	+22.8%	2.78	Spike → recovery
MSFT	**ContDn (IV > 1, DOWN)**	69	-11.4%	+12.1%	3.58	Bounded bilateral
MSFT	**RevDn (IV < 1, DOWN)**	35	-10.4%	+22.8%	2.62	Spike → bounded
MSTR	**RevDn (IV < 1, DOWN)**	116	**-32.2%**	**+70.2%**	4.04	**Bilateral explosive**
MSTR	**ContUp (IV < 1, UP)**	116	**-34.8%**	**+82.5%**	3.26	**Bilateral explosive**

4.1 Bilateral Explosiveness Hierarchy

A clear hierarchy emerges in the magnitude of bilateral tail risk at long horizons:

- **MSTR** exhibits the most extreme bilateral tails, with downside risk exceeding -30% and upside potential exceeding +70% to +80% in both supported regimes.
- **TSLA** displays strong bilateral explosiveness only in the put-dominated downside regime (ContDn), with -38% downside and +36% upside tails.
- **MSFT** shows bounded bilateral behavior across all regimes, with terminal tail magnitudes confined to approximately ±10–23%.
- **SPY** exhibits the strongest suppression of bilateral tail risk, with terminal tails limited to approximately ±11–23% even under extreme IV regimes.

This ordering is stable across horizons.

4.2 Kurtosis as a Mechanism Indicator

Kurtosis differentiates between transient dispersion and sustained fat-tailed behavior:

- **Extreme short-horizon spikes (K > 8)** occur following downside shocks in call-dominated regimes:
  - TSLA ContDn (K = 8.55 at lag 1)
  - SPY RevDn (K = 14.22 at lag 1)
  - MSFT RevDn (K = 8.44 at lag 1)
- **Sustained elevated kurtosis (K ~ 3.5–4.5)** accompanies persistent bilateral explosiveness:
  - MSTR in both supported regimes
  - TSLA ContDn at longer horizons
- **Near-Gaussian kurtosis (K < 3)** corresponds to drift-dominated tail expansion:
  - SPY ContDn
  - MSFT ContDn
  - TSLA RevDn

Importantly, large tail magnitudes may persist even as kurtosis compresses, indicating a transition from rare-event dominance to broader distributional widening.

4.3 IV-Ratio Regime Effects

IV > 1 (Puts Expensive) + DOWN Shock → ContDn

- TSLA: **-38% / +36%** — bilateral explosive
- MSFT: -11% / +12% — bounded bilateral
- SPY: -11% / +12% — benign bilateral
- MSTR: **structurally absent**

Put-dominated downside regimes generate bilateral explosion only for convex single-names. Mega-caps remain bounded, while index structure suppresses convexity entirely.

IV < 1 (Calls Expensive) + DOWN Shock → RevDn

- TSLA: -16% / +22% — asymmetric, upside-biased
- MSFT: -10% / +23% — transient spike, recovery
- SPY: -11% / +23% — extreme initial spike, rapid normalization
- MSTR: **-32% / +70%** — bilateral explosive

Call-dominated downside regimes commonly produce short-horizon kurtosis spikes that normalize quickly, except in MSTR where structural convexity persists.

4.4 Asset Structure Governs Bilateral Feasibility

Asset structure determines whether bilateral explosive tails are feasible:

- **Leveraged proxy (MSTR):** Persistent bilateral explosiveness in all supported regimes.
- **Volatile single-name (TSLA):** Bilateral explosiveness only when puts are expensive after downside shocks.
- **Mega-cap (MSFT):** Bounded bilateral tails across all regimes.
- **Broad index (SPY):** Structural suppression of sustained bilateral explosiveness.

The IV ratio therefore functions as a regime selector for tail geometry, while asset structure determines whether bilateral convexity can manifest.

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5. Interpretation and Mechanisms

5.1 The IV Ratio as a Bilateral Tail Selector

Core finding: The Put–Call IV ratio does not predict return direction. Instead, it selects which conditional forward return distributions exhibit **bilateral explosive tails**, which display **asymmetric tail structures**, and which remain **bounded and near-Gaussian**.

When puts are expensive (IV > 1)

After DOWN shocks (ContDn scenario)

- **Convex single-names (TSLA):**
  Strong bilateral explosion, with terminal tails of approximately **-38% / +36%**. Both downside crash risk and upside rebound potential are extreme.
- **Mega-cap (MSFT):**
  Bounded bilateral behavior, with tails confined to roughly **-11% / +12%**.
- **Index (SPY):**
  Benign bilateral behavior, also confined to approximately **-11% / +12%**, despite occasional short-horizon kurtosis spikes.
- **Leveraged proxy (MSTR):**
  Structurally absent (N = 0) — this IV regime does not occur in sample.

Mechanism.
When puts are expensive following downside shocks, option markets are not pricing directional bearishness. They are pricing **two-sided uncertainty**. In convex assets such as TSLA, this uncertainty manifests as true bilateral tail explosion: the asset may crash further or rally violently with comparable tail risk. In contrast, mega-cap diversification (MSFT) and index structure (SPY) suppress the feasibility of sustained bilateral convexity.

After UP shocks (RevUp scenario) When puts are expensive following upside shocks, forward return distributions do not exhibit bilateral explosiveness.

- **TSLA:**
  Rare regime (N ~ 5) with thin-tailed behavior. Downside risk dominates (~ -24%) while the upside tail remains modest (~ +6%), and kurtosis is sub-Gaussian.
- **MSFT and SPY:**
  Moderate bilateral tails with bounded magnitudes (~ ±10–20%) and near-Gaussian kurtosis. No evidence of sustained convexity.
- **MSTR:**
  Structurally absent (N = 0). Puts never become relatively expensive following upside shocks.

Mechanism.
Put-dominated regimes following upside shocks reflect **hedging demand rather than convex uncertainty**. Market participants insure recent gains against drawdowns, producing asymmetric or bounded tail structures rather than bilateral explosion. This regime represents tail compression or convexity unwind, not volatility amplification.

When calls are expensive (IV < 1)

After DOWN shocks (RevDn scenario)

- **TSLA:**
  Asymmetric tail structure, with moderate downside risk (**-16%**) and larger upside potential (**+22%**), accompanied by thin tails.
- **MSFT:**
  Transient fat-tail spike followed by normalization, ending with bounded downside (**-10%**) and stronger upside recovery (**+23%**).
- **SPY:**
  Extreme initial kurtosis spike (**K = 14.22**) reflecting constituent dispersion, followed by rapid normalization and upside recovery (**+23%**).
- **MSTR:**
  Bilateral explosive, with terminal tails of approximately **-32% / +70%**, indicating persistent two-sided convexity.

Mechanism.
Call-dominated regimes following downside shocks typically reflect expectations of mean reversion. In indices and mega-caps, this expectation is delivered through short-lived dispersion and rapid convergence. In MSTR, however, leveraged exposure prevents normalization, producing sustained bilateral tail risk.

After UP shocks (ContUp scenario)

- **MSTR:**
  Bilateral explosive, with terminal tails of approximately **-35% / +82%**.
- **TSLA:**
  Bilateral expansion with upside bias (**-24% / +35%**) and near-Gaussian kurtosis.
- **MSFT and SPY:**
  Moderate bilateral tails, confined to roughly **±10–13%**.

Mechanism.
When calls remain expensive after upside shocks, convex assets continue to exhibit two-sided risk, while mega-cap and index structure constrain tail magnitudes.

5.2 Kurtosis Timing Reveals the Underlying Mechanism

Kurtosis distinguishes between **transient dispersion** and **persistent bilateral fat tails**. Three distinct patterns emerge:

1. Sustained Elevation (K $\geq$ 4 across multiple horizons)

- **TSLA ContDn:** $8.55 \rightarrow 2.43 \rightarrow 4.06$
- **MSTR RevDn and ContUp:** sustained **K** $\sim$ **3.3–4.5**

Interpretation:
Fat-tailed structure persists across horizons. Tail risk remains active in both directions. These scenarios represent genuine bilateral explosiveness.

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2. Spike → Rapid Normalization (K > 8 initially, K < 3 by lag 10)

- **SPY RevDn:** 14.22 → 5.94 → 2.56
- **MSFT RevDn:** 8.44 → 3.53 → 2.62

Interpretation:
Initial dispersion dominates short horizons, driven by heterogeneous constituent responses. Rapid convergence follows, preventing sustained bilateral explosion.

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3. Stable Low Kurtosis (K < 3 throughout)

- **SPY ContDn**
- **MSFT ContDn**
- **TSLA RevDn**

Interpretation:
Thin-tailed, near-Gaussian distributions. Tail expansion is drift-dominated rather than convexity-driven.

Importantly, large terminal tail magnitudes may still occur even when kurtosis compresses, reflecting a shift from rare-event dominance to broad distributional widening.

5.3 Why Bilateral Tails Matter

Traditional tail risk frameworks focus on a single tail, typically downside VaR or ES. This approach misses critical features documented here:

1. **Bilateral explosive regimes** where both tails are simultaneously extreme (e.g., TSLA ContDn: **-38% / +36%**).
2. **Asymmetric regimes** where upside recovery dominates downside risk (e.g., SPY and MSFT RevDn).
3. **Hedging paradoxes**, where put buyers face substantial upside risk at the same time as crash risk.

Economic implications:

- In TSLA ContDn, option markets price bilateral uncertainty, not directional bearishness.
- In SPY RevDn, extreme initial kurtosis reflects transient constituent dispersion rather than persistent risk.
- In MSTR, bilateral explosiveness across regimes reflects structural leverage rather than regime switching.

5.4 Structural Absence as Information

MSTR exhibits **N = 0** for the following scenarios across all horizons:

- **RevUp (IV > 1, UP shock)**
- **ContDn (IV > 1, DOWN shock)**

This absence is systematic and informative, not a sampling artifact.

Interpretation.
The leveraged Bitcoin exposure embedded in MSTR produces option pricing dynamics that eliminate certain IV regime and shock combinations. When MSTR rallies, puts never become relatively expensive. When MSTR falls, calls never dominate. The IV ratio therefore does not freely switch regimes for MSTR; instead, it reveals structural constraints imposed by the underlying exposure.

Structural absence itself becomes a signal about feasible return geometry.

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6. Practical Implications

6.1 Risk Management

1. Measure BOTH tails, not just downside.
This study shows that risk is often bilateral, not one-sided.

- **TSLA ContDn (IV > 1, DOWN shock):** -38% downside tail AND +36% upside tail
- Traditional downside-only VaR captures only half the uncertainty
- Bilateral tail measurement reveals the true risk geometry faced by portfolios

Ignoring the upper tail can materially misstate risk in convex assets.

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2. Kurtosis monitoring is essential.
Kurtosis timing distinguishes between benign dispersion and persistent convexity.

- **Extreme initial spikes (K > 8):**
  - TSLA ContDn: 8.55
  - SPY RevDn: 14.22
  - MSFT RevDn: 8.44
- **Interpretation:**
  - **Spike → rapid normalization (SPY, MSFT):** structural dispersion, no sustained risk
  - **Sustained elevation (TSLA, MSTR):** persistent bilateral tail exposure

Risk rule:
- Transient spikes (index, mega-cap) → monitoring only
- Sustained elevation (convex single-names) → hedge or reduce exposure

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3. Asset structure determines hedging strategy.

- **Index exposures (SPY):**
  Extreme initial kurtosis spikes but rapid normalization → short-dated hedges sufficient
- **Mega-caps (MSFT):**
  Moderate spikes with partial normalization → medium-dated hedges appropriate
- **Convex single-names (TSLA, MSTR):**
  Sustained bilateral explosiveness → long-dated hedges or position size reduction required

Risk cannot be managed uniformly across asset classes.

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4. IV regime signals bilateral uncertainty, not direction.

- **TSLA ContDn (IV > 1, DOWN):** expect ±36–38% bilateral tails, not a directional crash
- **SPY RevDn (IV < 1, DOWN):** expect transient chaos followed by upside recovery
- Hedge ratios should be adjusted for **bilateral vs. asymmetric** tail structures

The IV ratio identifies the *shape* of risk, not the *direction* of returns.

6.2 Portfolio Construction

1. Diversification effects are non-linear.

- **SPY ContDn:** ±11–12% terminal tails
- **TSLA ContDn:** ±36–38% terminal tails

A portfolio of volatile single-names does **not** replicate index behavior.
Diversification suppresses bilateral explosiveness structurally, not just via correlation.

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2. Leverage amplifies bilateral risk.

- **MSTR:** ±32–35% downside and ±70–82% upside tails in both supported regimes
- Leveraged exposure creates symmetric convexity where both tails explode
- Position sizing must reflect **bilateral tail magnitudes**, not mean returns

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3. Multi-day horizon planning is critical.

- Tail magnitudes accumulate over **10–20 trading days**
- Daily risk metrics underestimate multi-day bilateral exposure
- Example: TSLA ContDn downside tail deepens from -8.9% (lag 1) to -38.4% (lag 20)

Risk unfolds over horizons relevant to real portfolios, not single-day VaR windows.

6.3 Options Trading

1. Put buying in ContDn scenarios (IV > 1, DOWN shock).

- **TSLA:** Buying puts for -38% downside protection also exposes to +36% upside tail
- **SPY/MSFT:** Buying puts for -11% protection is expensive given bounded bilateral tails

Directional put buying may be inefficient when uncertainty is bilateral.
Strategies that monetize two-sided risk (e.g., straddles, strangles) may be more appropriate.

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2. RevDn scenarios (IV < 1, DOWN shock).

- **SPY/MSFT:** Extreme initial dispersion normalizes quickly; upside recovery dominates
- Short-dated calls may capture rebound potential (~ +23% upper tail)

Mean reversion dominates in structurally diversified assets.

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3. Structural absence must be respected.

- **MSTR:** RevUp and ContDn regimes do not occur (N = 0)
- Certain option strategies are structurally infeasible
- Do not wait for IV regime transitions that cannot happen

Structural absence is itself actionable information.

6.4 Alpha Generation vs. Risk Measurement

Critical distinction: This analysis is diagnostic, not predictive.

- We condition on realized forward paths to measure tail geometry
- We do NOT predict which tail will realize ex ante
- The IV ratio identifies **which risk distribution is active**, not **which outcome will occur**

Practical uses:

- **Risk budgeting:** Allocate capital based on scenario-specific bilateral tail magnitudes
- **Stress testing:** Use conditional distributions (e.g., TSLA ContDn ±36–38%), not generic historical VaR
- **Hedge selection:** Match hedge structures to bilateral vs. asymmetric tail geometry

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Not recommended:

- Directional trading based on IV ratio alone
- Assuming "puts expensive = stock goes down"
- Ignoring upside tail risk when hedging downside exposure

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7. Limitations and Caveats

7.1 Conditional (Ex Post) Analysis

All results are conditional on information observed at time t (shock direction and IV ratio regime), but forward return distributions are measured ex post. The analysis does not claim the ability to predict which tail will realize, nor to forecast returns directionally.

The objective is diagnostic rather than predictive: to characterize how the geometry of forward return distributions differs across conditioning regimes. The IV ratio is treated as a state variable governing tail structure, not as a trading signal.

7.2 Uneven and Scenario-Specific Sample Sizes

Sample sizes vary materially across scenarios and assets:

- **TSLA RevUp (IV > 1, UP shock):** N = 5 — rare regime; results are suggestive only
- **TSLA RevDn (IV < 1, DOWN shock):** N ~ 19 — moderate sample
- **TSLA ContDn (IV > 1, DOWN shock):** N ~ 83 — large, stable sample
- **MSTR RevDn / ContUp:** N ~ 116–120 — large samples
- **MSTR RevUp / ContDn:** N = 0 — structurally absent regimes

Small-sample scenarios are reported for completeness but are not over-interpreted. Structural absence (N = 0) is treated as economically meaningful rather than as missing data.

7.3 Threshold Choice for Shock Definition

Large shocks are defined using 80/20 percentile thresholds of daily returns. This choice balances statistical power with the desire to isolate genuinely large moves.

Alternative thresholds would alter sample sizes and tail magnitudes: - Tighter thresholds (e.g., 90/10) reduce power - Looser thresholds (e.g., 70/30) dilute the notion of a large shock

The qualitative findings—bilateral explosiveness versus bounded or asymmetric tails—are not driven by marginal threshold tuning, but magnitudes may vary.

7.4 Interpretation of Kurtosis

Kurtosis is used as a reduced-form diagnostic of tail heaviness and distributional shape. High kurtosis indicates fat-tailed behavior but does not identify the underlying structural drivers.

Potential contributors include: - Option market gamma exposure - Volatility feedback effects - Balance-sheet leverage - Liquidity constraints - Underlying asset convexity

This study intentionally measures outcomes (tail geometry), not mechanisms at the microstructure level.

7.5 Asset Coverage and Generalizability

The four assets studied span distinct structural categories: - Broad index (SPY) - Mega-cap equity (MSFT) - Volatile single-name (TSLA) - Leveraged proxy (MSTR)

While these categories capture key equity archetypes, results for any individual asset—particularly TSLA and MSTR—should not be assumed to generalize to all equities. MSTR, in particular, reflects idiosyncratic exposure to Bitcoin and is not representative of typical single-name behavior.

7.6 Sample Period Dependence

Each asset is analyzed over an approximately 600-day lookback window. Results are therefore conditional on the prevailing market structure during the sample period.

Longer samples or different market regimes (e.g., prolonged low-volatility environments) may alter the frequency or intensity of certain scenarios, though structural distinctions across asset types are expected to persist.

7.7 Implied Volatility Construction

The IV ratio is constructed using 30-day maturity options and ten strikes surrounding at-the-money. The analysis does not explore: - Term structure effects across maturities - Deep in- or out-of-the-money skew dynamics - Alternative IV ratio constructions

Different IV definitions may shift regime classifications at the margin but are unlikely to overturn the core finding that the IV ratio governs tail geometry rather than directional outcomes.

Summary.
These limitations do not undermine the central conclusion: the Put–Call IV ratio functions as a selector of conditional return geometry, not as a directional predictor. Asset structure ultimately determines whether bilateral explosive tails are feasible, bounded, or structurally suppressed.

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8. Conclusions

This study shows that the Put–Call implied volatility ratio does not function as a directional predictor of returns. Instead, it operates as a state variable that governs the **geometry of conditional forward return distributions**, determining whether risk resolves through bilateral explosiveness, asymmetric tail behavior, or bounded near-Gaussian dynamics.

8.1 The IV Ratio Selects Bilateral Tail Geometry, Not Direction

Across assets, the IV ratio consistently selects *how* tails resolve, not *which direction* returns take.

When **puts are expensive (IV > 1) following a DOWN shock** (ContDn scenario):

- **TSLA:**
  Bilateral explosive tails, with ES_5 ~ -38% and ES_95 ~ +36% at a 20-day horizon, accompanied by elevated kurtosis that spikes early (K = 8.55) and re-elevates at longer horizons (K ~ 4.06).
- **MSFT:**
  Bounded bilateral behavior, with tails confined to approximately -11% / +12% and near-Gaussian kurtosis.
- **SPY:**
  Benign bilateral behavior, also confined to approximately -11% / +12%, despite occasional extreme short-horizon kurtosis spikes.
- **MSTR:**
  Structurally absent — this IV regime does not occur in sample.

These results demonstrate that put-dominated regimes do not signal directional bearishness. In convex single-names, they signal **bilateral uncertainty**, while asset structure determines whether that uncertainty can amplify explosively.

8.2 Kurtosis Timing Discriminates Risk Mechanisms

Kurtosis provides critical information about the *mechanism* driving tail behavior.

- **Sustained elevation (K ≥ 4 across horizons):**
  Indicates persistent bilateral fat tails. This pattern characterizes **TSLA ContDn** and **MSTR in all supported regimes**, reflecting true bilateral explosiveness.
- **Spike followed by rapid normalization (K > 8 → K < 3 within ~10 days):**
  Reflects transient dispersion and rapid convergence, as seen in **SPY RevDn** and **MSFT RevDn**. These spikes are dramatic but economically benign.
- **Stable low kurtosis (K < 3):**
  Indicates thin-tailed, drift-dominated dynamics, characteristic of **SPY ContDn**, **MSFT ContDn**, and **TSLA RevDn**.

Thus, the *timing* of kurtosis normalization is as important as its magnitude.

8.3 Asset Structure Governs Bilateral Feasibility

Whether bilateral explosive tails can occur is determined primarily by asset structure:

- **Leveraged proxy (MSTR):**
  Exhibits extreme bilateral explosiveness in all supported regimes, with terminal tails reaching approximately -32% to -35% on the downside and +70% to +82% on the upside.
- **Volatile single-name (TSLA):**
  Exhibits bilateral explosiveness only in specific regimes, most notably when puts are expensive after downside shocks.
- **Mega-cap (MSFT):**
  Displays bounded bilateral behavior across all regimes.

- **Broad index (SPY):**
  Structurally suppresses sustained bilateral explosiveness, even when short-horizon dispersion is extreme.

Diversification and balance-sheet structure therefore act as hard constraints on tail amplification.

8.4 Measuring Both Tails Is Essential

Traditional risk frameworks that focus on a single tail (typically downside VaR or ES) miss critical features documented here:

- **Bilateral explosive regimes**, where crash risk and rally risk are simultaneously extreme (e.g., TSLA ContDn).
- **Asymmetric recovery regimes**, where upside tails dominate downside risk (e.g., SPY and MSFT RevDn).
- **Hedging paradoxes**, where downside protection coincides with substantial upside risk.

Accurate risk assessment requires explicit measurement of **both tails**.

8.5 Initial Kurtosis Spikes Have Opposite Meanings Depending on Persistence

Large initial kurtosis spikes (>8) are not uniformly dangerous:

- **Rapid normalization** (e.g., SPY RevDn: $K = 14.22 \rightarrow \sim 2.8$) signals transient chaos with benign resolution.
- **Slow normalization or re-elevation** (e.g., TSLA ContDn: $K = 8.55 \rightarrow 2.43 \rightarrow 4.06$) signals sustained bilateral tail risk.

Failing to distinguish between these cases can lead to systematic over- or under-hedging.

8.6 Structural Absence Is Informative

MSTR exhibits $\mathbf{N = 0}$ for the following scenarios across all horizons:

- **RevUp (IV > 1, UP shock)**
- **ContDn (IV > 1, DOWN shock)**

This absence is not noise but reflects structural constraints imposed by leveraged Bitcoin exposure. Certain IV regime and shock combinations are infeasible, and the absence of observations itself conveys information about market pricing of volatility.

8.7 Practical Implications

The findings have direct implications for risk management and portfolio construction:

- **Risk budgeting:** Use scenario-specific bilateral tail distributions rather than generic historical VaR.
- **Hedging:** Match hedge structures to bilateral versus asymmetric tail geometry.
- **Portfolio construction:** Recognize that index diversification suppresses bilateral explosiveness non-linearly.
- **Options strategy:** In convex assets with IV > 1 after downside shocks, bilateral uncertainty strategies may dominate directional hedges.

8.8 Central Takeaway

The most actionable insight is not predictive but diagnostic:

When a **convex single-name equity such as TSLA experiences a large downside shock and puts become expensive**, the forward return distribution enters a regime of **extreme bilateral uncertainty**, with tail magnitudes on the order of ±35–40% over a 20-day horizon. This is not a signal of an impending crash; it is a signal that **both crash and violent rebound are simultaneously plausible**.

Risk management frameworks that focus only on downside exposure fundamentally mischaracterize the nature of risk in such regimes.

Data Availability

All empirical results reported in this study are derived from daily equity returns and option-implied volatility data constructed by OVVO Labs. To preserve readability in the main text, condensed tables are presented for each asset and scenario. Complete bilateral tail statistics — including Value-at-Risk, Expected Shortfall, kurtosis, and sample sizes across all horizons (1–20 trading days) and conditioning regimes — are provided in full in the accompanying online repository:

https://github.com/OVVO-Financial/Finance/blob/main/Data/Put%20Call%20IV%20Results

All figures and interpretations in the paper are based on these full results tables.

References

Implied volatility data: OVVO Labs options web application (30-day maturity, ATM ±5 strikes)

Statistical methods: Nonparametric percentile estimation, excess kurtosis calculation in R

Assets: TSLA, MSFT, SPY, MSTR daily data (~600 day lookback per asset)